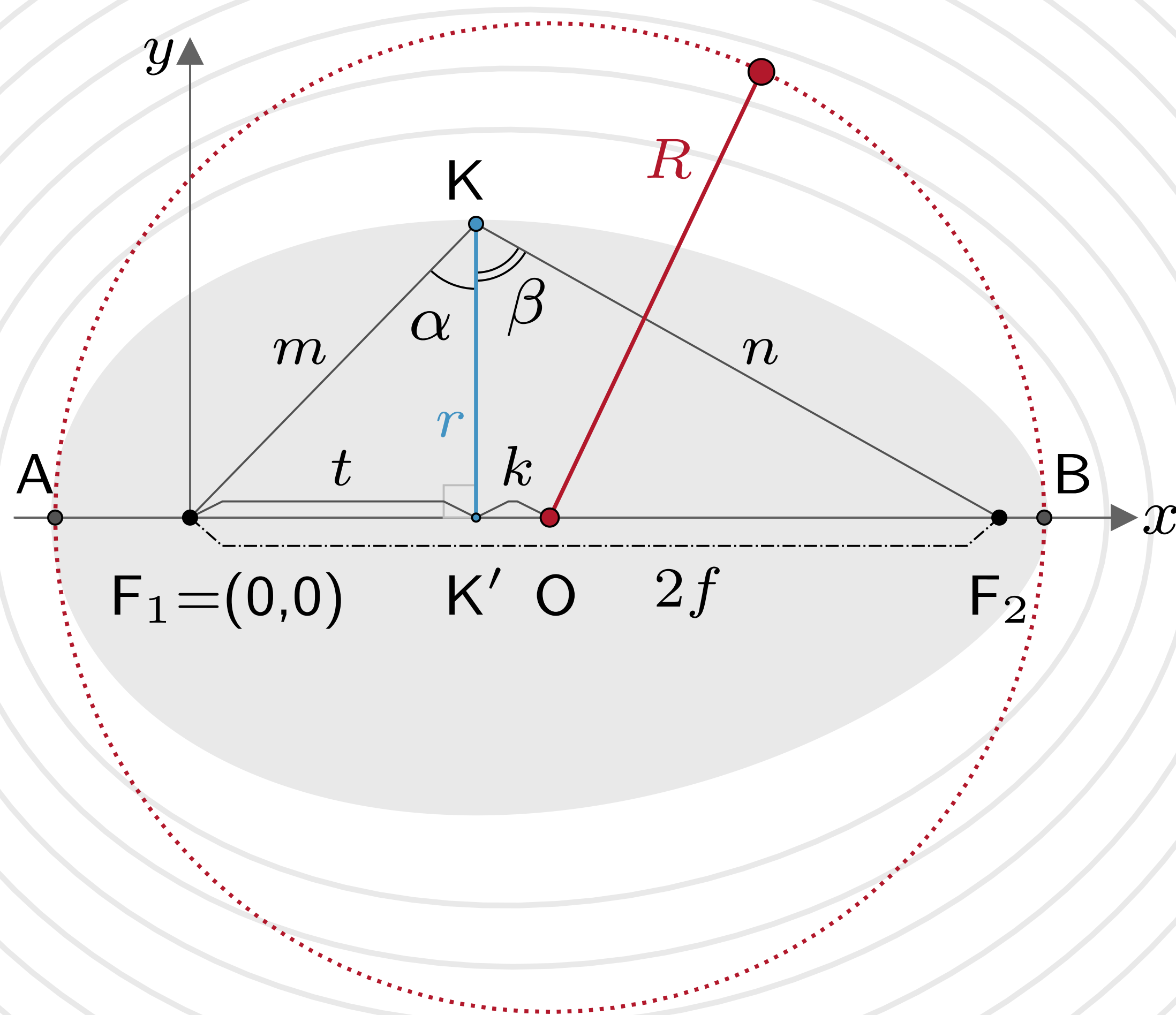




Main Contributions:

- the egg-shape study relatively to its Smallest Enclosing Circle (SEC);
- the discovery of the geometric properties that explain the foci locations through the weight μ ;
- the discovery of the connection between the focal distance ($2f$), sum of the weighted distances ($2c$), $\sin \beta$ and τ :

$$\tau \stackrel{\text{def}}{=} \cos(\beta - \alpha) = \frac{c \sin \beta}{f}$$
- the derivation of the egg-shape parameters through SEC and largest inscribed circle;
- the introduction of the **linear** (with a potential parallel execution) **algorithm** to find the egg-shape parameters given a binary image.

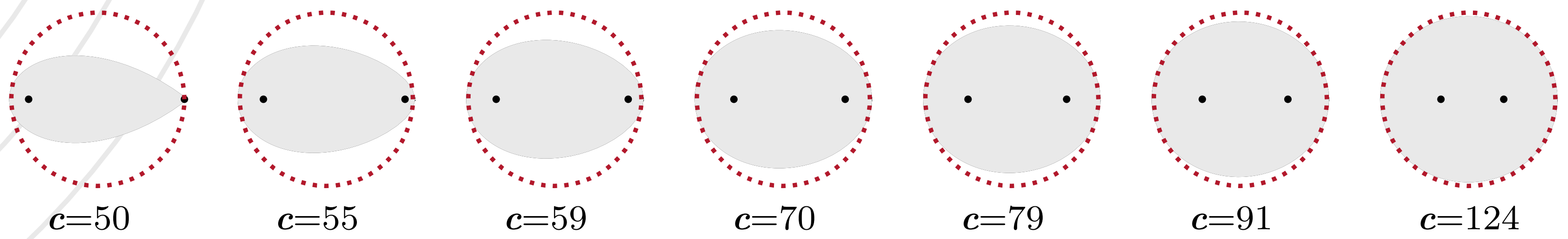
What is the Egg-shape?

- generalization of an ellipse [Gab22];
- set of points with the same weighted sum of the distances to two foci, F_1 and F_2 :

$$d(P, F_1) + \mu d(P, F_2) = 2c,$$
 where $\mu \in (0, 1]$ is the weight of F_2
- smooth curve of a circular, elliptic or oval shape that could contain a sharp corner;
- one symmetry axis.

Why the Egg-shape?

- expresses a wide variety of real-world objects, e.g. grape, oyster, human face, tennis racket and dirigible [HK24];
- describes **elongation**, **orientation** and **narrowing**;
- requires one extra parameter (μ) compared to the ellipse;
- expresses **deformation** and has a connection to **physical phenomena**.

for $f=50$, $\mu=0.8$:

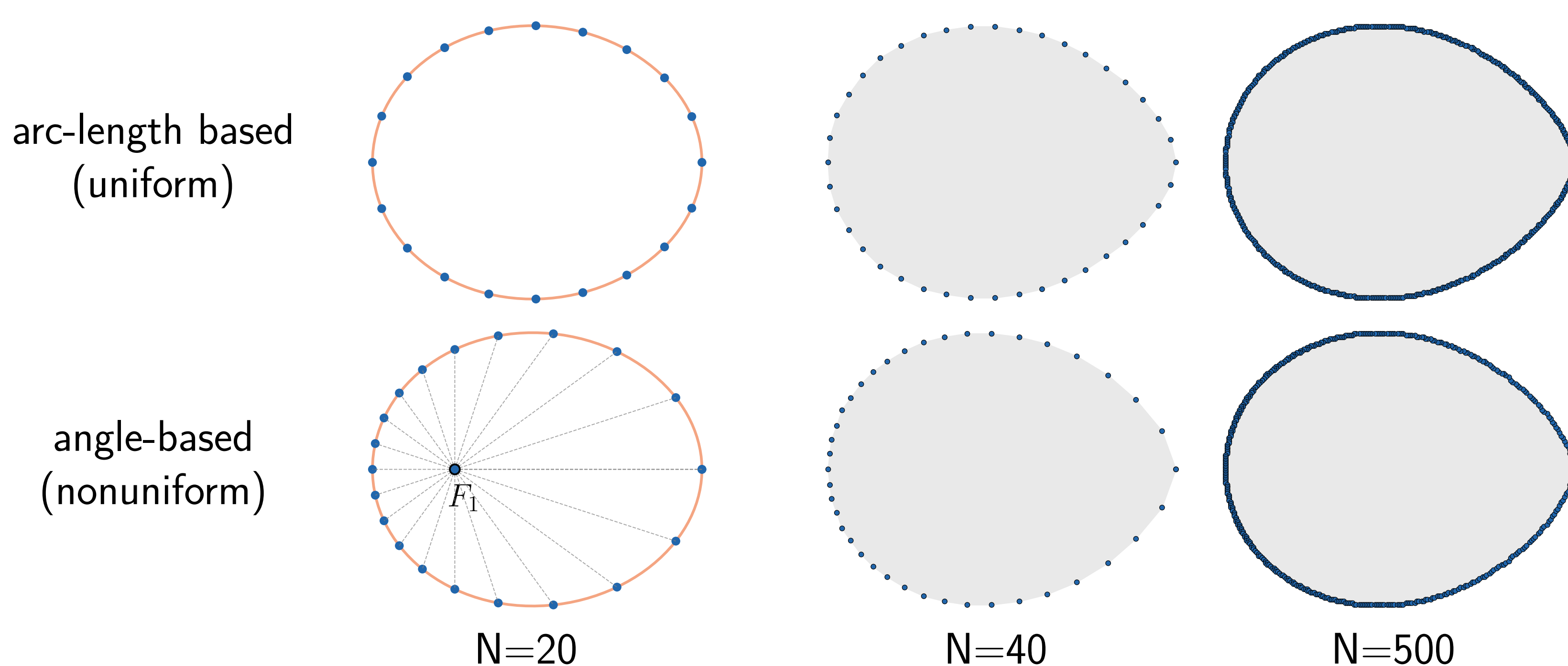
Nomenclature

$F_1 = (0, 0)$ $F_2 = (2f, 0)$	Focal points	$2f$	Focal length, $d(F_1, F_2)$
μ	Weight of F_2 , $\mu \in (0, 1]$	$2c$	Weighted sum of the distances to foci
A, B	Endpoints of the major axis	R	Radius of SEC
$K = (t, r)$	y-axis extremum of the upper half of the egg-shape	$K' = (t, 0)$	Projection of K onto $\overline{F_1 F_2}$; centre of the largest inscribed circle
α	$\angle K' K F_1$	β	$\angle K' K F_2$
τ	$\cos(\beta - \alpha)$	O	SEC centre
k	Distance between O and K'	r	Radius of the largest inscribed circle

Egg-shape Parameter Finding: Comparison to the State-of-the-Art

	$ \mu - \mu_{gt} $	N=40				N=500			
		nonuniform	uniform	circular	oval	nonuniform	uniform	circular	oval
our	≤ 0.01	34.6%	42.2%	20.4%	48.9%	51.7%	54.8%	25.8%	69.4%
	≤ 0.1	87.1%	86.7%	67.2%	98.7%	91.3%	91.2%	76.5%	99.8%
[HK24]	≤ 0.01	30.9%	34.4%	6.1%	48.2%	41.6%	48.5%	7.8%	66.9%
	≤ 0.1	74.5%	73.8%	35.2%	97%	67.4%	79.2%	37.3%	94.5%

Sampling Types [HK24]



Test Dataset:

- 6272 binary images of size 512×512 containing the egg-shapes:
 - 1568 combinations of the egg-shape parameters for $R = 100$;
 - uniform and nonuniform samplings;
 - $N = 40$ and $N = 500$.
- 37% of the dataset contains nearly circular shapes.

Summary of the Results:

- the proposed algorithm rests on the exact geometric solution and SEC; the state-of-the-art method [HK24] assumes the egg-shape and the best-fitting ellipse to have the same elongation, defined as $\epsilon \approx \frac{1}{\varsigma} \mu^{\varsigma}$, where $\varsigma = \frac{2c}{f}$;
- the egg-shapes that have the sharp corner ($c = f$) or resemble an oval lead to more accurate results due to unambiguous form;
- subtle differences between the circular egg-shapes are hardly visible to human and are challenging for both algorithms;
- the percentage of the cases, when the accuracy of finding μ is below the specified threshold, is higher for the proposed approach in all the observed scenarios.

References:

- [Gab22] Aysylu Gabdulkhakova. *From Circles to Generalized Conics: Enriching the Properties of 2D Shape Representation and Description*. Tech. rep. PRIP-TR-156, extended version of PhD thesis. Pattern Recognition and Image Processing Group, Technische Universität Wien, 2022.
- [HK24] Jiří Hladůvka and Walter G. Kropatsch. "Fitting Egg-Shapes to Discretized Object Boundaries". In: *Discrete Geometry and Mathematical Morphology (DGMM 2024), LNCS, vol. 14605*. Ed. by Sara Brunetti, Andrea Frosini, and Simone Rinaldi. Cham: Springer Nature Switzerland, 2024, pp. 107–119. DOI: 10.1007/978-3-031-57793-2_9.